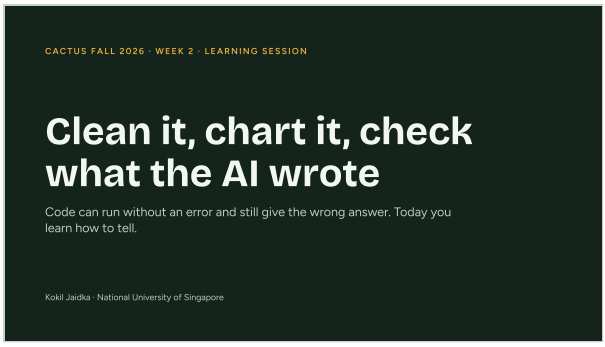
# CACTUS Week 2: Clean It, Chart It




## Slide 1

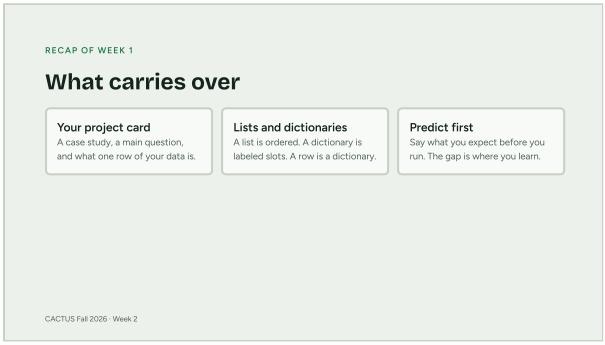
**TIME** 1 min (0:00). This is a 60-minute session; the coding session afterwards is where people practice.

**SAY** Welcome back. This week is about AI as pair programmer: you read it, test it, and do not trust it blindly.

**DO** Welcome people by name where you can. Post the course site link. Say the plan: one hour now, then a short practice page, then the two-hour coding session.

**NOTE** Types and messy data are in the required reading (textbook Chapter 5) and in the appendix slides at the end. We do not present them today.

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## Slide 2

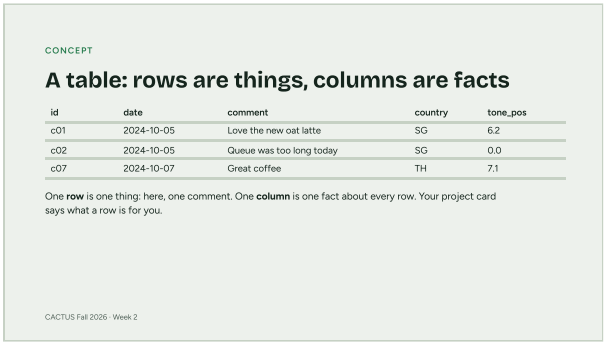
**TIME** 3 min (0:01)

**SAY** Quick recap. Your project card. Lists and dictionaries: a row of data is a dictionary. And the habit: predict before you run.

**DO** Ask one or two people to say their case study in one sentence. Keep it short.

**WATCH FOR** Time. This slide is a warm-up, not a discussion.

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## Slide 3

**TIME** 5 min (0:04)

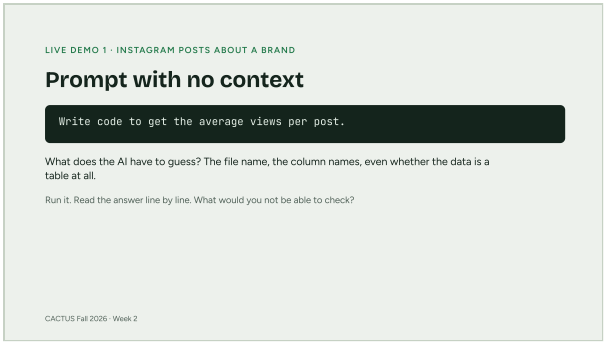
**SAY** A table: rows are things, columns are facts about them. Here one row is one comment. Your project card says what a row is for you. Two things to know before we start: a value can look the same and behave differently (a number stored as text, for example), and real data is messy, with missing values, duplicates, and uneven groups. You read about both in Chapter 5; the appendix slides have a short version.

**DO** Ask the room: what is one row in your project? Take two answers.

**IF ASKED "What is pandas?"** A Python tool for working with tables. Chapter 5 introduces it.

**WATCH FOR** Do not teach types or missing values in depth. Point to the reading and move on.

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## Slide 4

**TIME** 6 min (0:09 to 0:15)

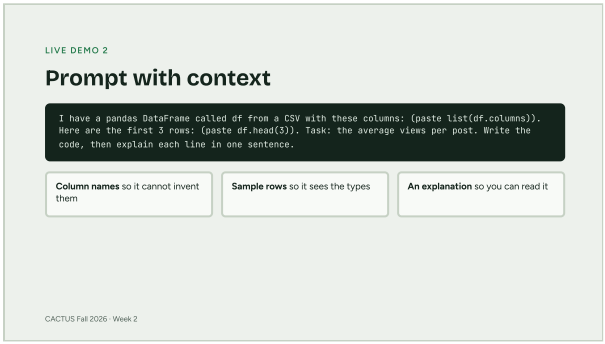
**DO (live demo)** Open the Lululemon file (lululemon_instagram.csv) in Colab, share your screen, and open an AI chat. Paste ONLY the prompt on the slide: "Write code to get the average views per post." Do not mention the file, columns, or any context.

**SAY** Watch what the AI has to guess: the file name, the column names, even whether this is a table.

**DO** Read its answer for one or two minutes. Ask: which names did it make up? What would you not be able to check?

**WATCH FOR** Do not skip to the good prompt. Let the failure be uncomfortable for a moment. That is the lesson. Keep it short.

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## Slide 5

**TIME** 9 min (0:15 to 0:24)

**DO (live demo)** In the notebook, run df.columns and df.head(3) on the Lululemon data. Copy the output into the prompt on the slide and submit it. Compare with the first attempt.

**SAY** Context removed the guessing. Column names so it cannot invent them. Sample rows so it sees the types. An explanation so you can read it.

**DO** Read the new code once, line by line: what does this do to my data? Run it. Do not reveal the average yet; the next slide asks for a prediction.

**SAY (reviewer's rule)** If you cannot explain a line, reject it.

**WATCH FOR** Never paste private data into an AI chat. This dataset is anonymized, but say it as a habit.

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## Slide 6

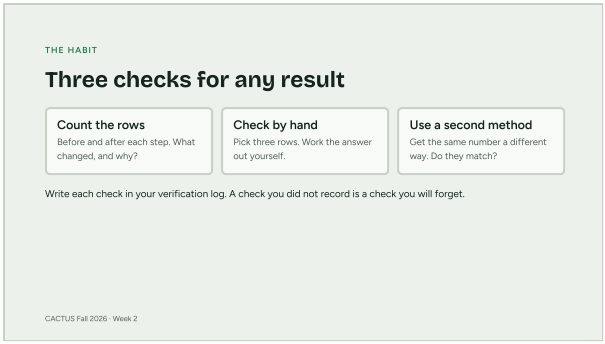
**TIME** 10 min (0:24 to 0:34)

**DO** Ask the room to predict the average views, then run it. It prints about 2,506. Then ask: how many posts have 0 views? Run (df["Views"] == 0).mean(). It shows about 91 percent.

**SAY** The code is correct. The question was wrong. In this data a zero often means not recorded, not that nobody viewed it. An average over mostly zeros does not describe what we meant.

**SAY** Running without error is not the same as being right. That is the lesson of the week.

**WATCH FOR** Do not make the AI the villain. The AI did what we asked. We did not check the data first.




## Slide 7

**TIME** 6 min (0:34 to 0:40)

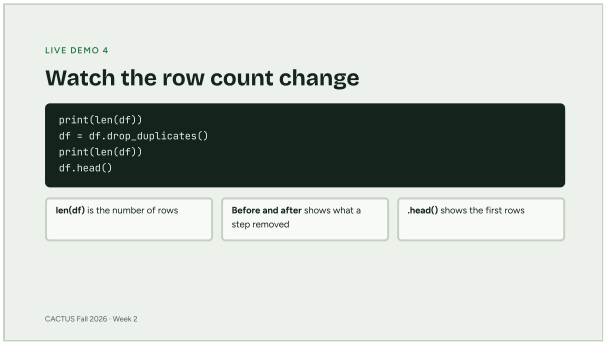
**SAY** Three checks for any result. Count the rows before and after each step. Check a few rows by hand. Get the same number a second way.

**DO** Give a one-line example from the Lululemon demo for each: row count (len before and after), by hand (look at three posts), second method (compare the mean with the count of zeros).

**SAY** Write each check in your verification log. A check you did not record is a check you will forget.

**WATCH FOR** People who think they must do all three every time. Say: pick the cheapest one that could catch the mistake you are worried about.

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## Slide 8

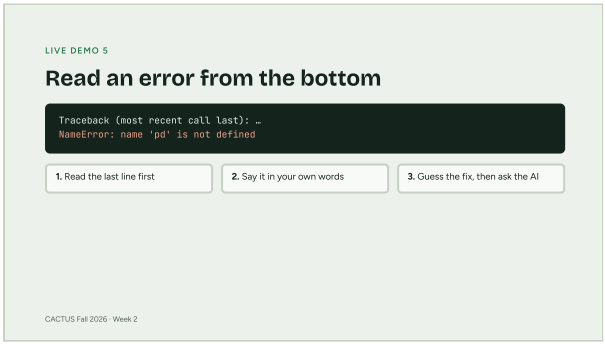
**TIME** 5 min (0:40 to 0:45)

**DO (live demo)** Run print(len(df)), then df = df.drop_duplicates(), then print(len(df)). Ask the room to predict the second number first. Then run df.head().

**SAY** len is the number of rows. Before and after shows what a step removed. head shows the first rows. This is the most useful ten seconds of debugging you will learn this week.

**IF THE COUNT DOES NOT CHANGE** say, good, that tells us there were no exact duplicates, and move on.

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## Slide 9

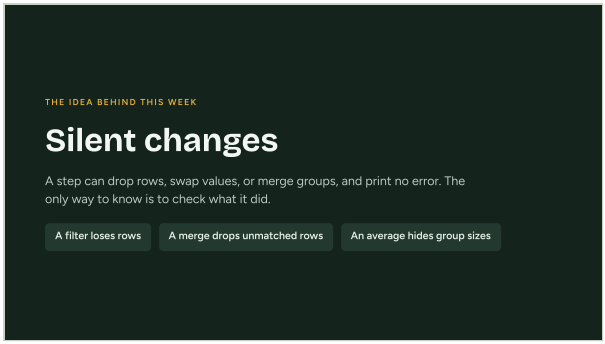
**TIME** 9 min (0:45 to 0:54)

**DO (live demo)** In a fresh notebook, run a cell that uses pd before you run the import cell. You get: NameError: name 'pd' is not defined. Do not explain it yet.

**SAY** Three steps. Read the last line first. Say it in your own words. Guess the fix, then ask the AI.

**DO** Ask a volunteer to say the last line in their own words (Python does not know the name pd yet). Paste the full message into an AI and compare its explanation with the volunteer's. Apply the fix and say in one sentence why it works.

**WATCH FOR** People panicking at red text. Say: errors are information, and the last line is the most useful part.




## Slide 10

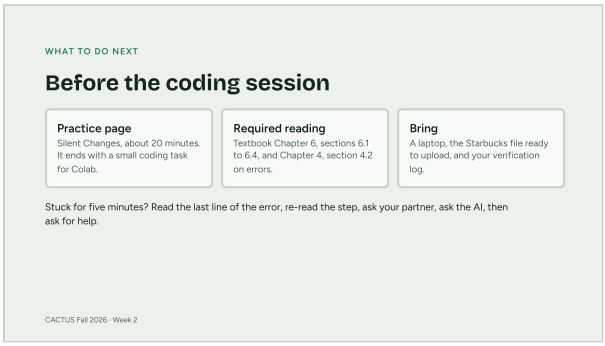
**TIME** 3 min (0:54)

**SAY** This is the idea behind the practice page. A step can drop rows, swap values, or merge groups and print no error. The only way to know is to check what it did.

**DO** Name the three examples on the slide. Say: the practice page has you build a cleaning pipeline and watch the row count change, then hunt three bugs that run without errors.

**NOTE** The practice page uses bug types that are different from the live exercise, so it does not give that away. Do not describe the live exercise bugs here.

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## Slide 11

**TIME** 3 min (0:57 to 1:00)

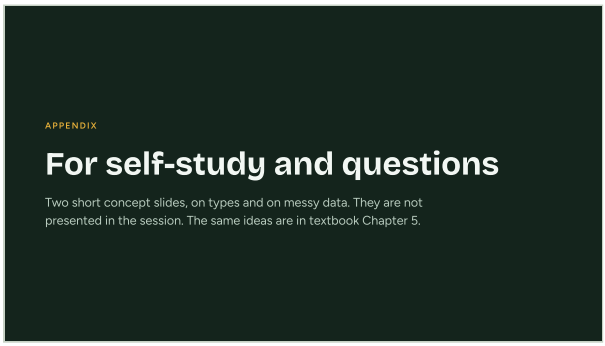
**SAY** Before the coding session: the practice page (about 20 minutes), the required reading (textbook Chapter 6 sections 6.1 to 6.4 and Chapter 4 section 4.2), and bring your laptop and the Starbucks file ready to upload.

**DO** Post the practice page and the course site link in the chat. Read out the stuck protocol: five minutes, read the last line, re-read the step, ask your partner, ask the AI, then ask for help.

**SAY** In the coding session there is no lecture. You work in pairs in breakout rooms and facilitators visit.

**CLOSE** Homework is to finish Exercise 5 on your own data and add one line to your project card on what could go wrong in your data. Thank everyone.

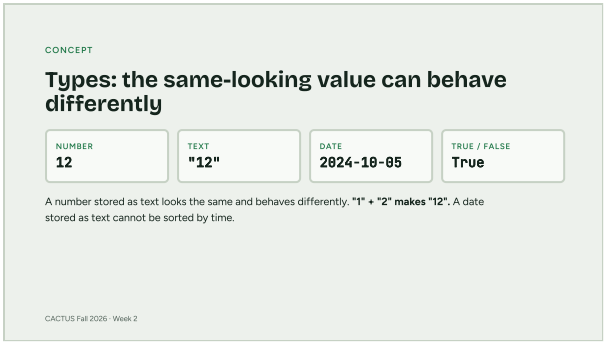
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## Slide 12

BACKUP SLIDES. Not presented in the session.

**NOTE** Use them if someone asks about data types or messy data, or point people to them for self-study. The same material is in textbook Chapter 5.




## Slide 13

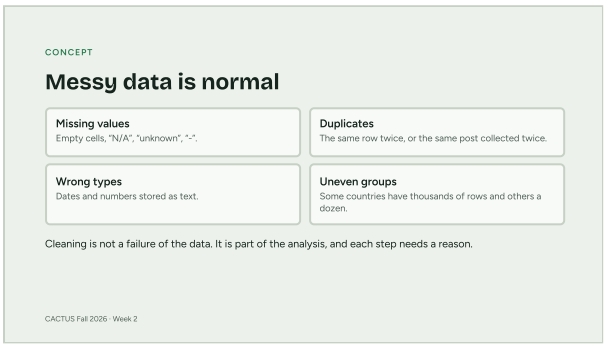
BACKUP SLIDE (appendix, not presented).

**SAY** The same-looking value can behave differently. 12 as a number, "12" as text, a date, true or false. Remember the puzzle page: quotes made a value text.

**DO** Point at 12 and "12". Say: add one and one as text and you get 11, not 2. In real data this happens silently when one bad cell, like "unknown", turns a whole column into text.

**IF ASKED how to check** df.info() shows each column's type. We use it in the notebook.

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## Slide 14

BACKUP SLIDE (appendix, not presented).

**SAY** Messy data is normal. Missing values, duplicates, wrong types, uneven groups. Cleaning is part of the analysis, and each step needs a reason you can say out loud.

**DO** Read the four cards. Say: you will see all four in the practice page and in the notebook.

**WATCH FOR** People who feel the data is wrong. Say: it is not wrong, it is real.

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